Develop a Sequence-to-Sequence model for translating English sentences into parts-of-speech tags

To develop a Sequence-to-Sequence neural network that converts English sentences into their corresponding POS-tag sequences.

Algorithm
1. Import TensorFlow and the required Keras preprocessing libraries.
2. Define English sentences and their corresponding POS-tag sequences.
3. Create separate tokenizers for English words and POS tags.
4. Convert both sets of sentences into integer sequences.
5. Pad the sequences to a fixed maximum length.
6. Create an Embedding layer to represent English words.
7. Add an LSTM encoder to understand the input sentence.

8. Use RepeatVector to transfer the encoded representation to the decoder.
9. Add an LSTM decoder with sequence output enabled.
10. Add a TimeDistributed Dense layer to predict POS tags at each time step.
11. Compile and train the model using the paired sequences.
12. Predict POS tags for a sample English sentence and display the output.

Program

```
import numpy as np
import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
tf.keras.utils.set_random_seed(32)
# English sentences
english = [
"i like apples",
"you read books",
"we play football",
"they eat rice",
"birds fly high",
"she writes letters",
"he drinks water",
"students learn coding"
]
# Corresponding POS tags
pos = [
"PRON VERB NOUN",
"PRON VERB NOUN",
"PRON VERB NOUN",
"PRON VERB NOUN",
"NOUN VERB ADV",
"PRON VERB NOUN",
"PRON VERB NOUN",
"NOUN VERB NOUN"
]
MAX_LEN = 3
# Tokenization
src_tok = Tokenizer(filters="")
tgt_tok = Tokenizer(

filters="",
lower=False
)
src_tok.fit_on_texts(english)
tgt_tok.fit_on_texts(pos)
# Convert sequences
X = pad_sequences(
src_tok.texts_to_sequences(english),
maxlen=MAX_LEN,
padding='post'
)
Y = pad_sequences(
tgt_tok.texts_to_sequences(pos),
maxlen=MAX_LEN,
padding='post'
```

```
)
src_vocab = len(src_tok.word_index) + 1
tgt_vocab = len(tgt_tok.word_index) + 1
# Build Seq2Seq model
model = models.Sequential([
layers.Embedding(src_vocab, 32),
layers.LSTM(64),
layers.RepeatVector(MAX_LEN),
layers.LSTM(64, return_sequences=True),
layers.TimeDistributed(
layers.Dense(tgt_vocab, activation='softmax')
)
])
# Compile
model.compile(
optimizer='adam',
loss='sparse_categorical_crossentropy',
metrics=['accuracy']
)
# Train
model.fit(
X, Y[..., None],
epochs=300,
verbose=0

)
# Test sentence
sentence = "i like apples"
sample = pad_sequences(
src_tok.texts_to_sequences([sentence]),
maxlen=MAX_LEN,
padding='post'
)
# Predict
pred = model.predict(
sample,
verbose=0
).argmax(axis=-1)[0]
id_to_pos = {
v: k for k, v in tgt_tok.word_index.items()
}
print("English Sentence:", sentence)
print(
"POS Translation:",
" ".join(id_to_pos.get(int(i), "PAD") for i in pred)
)
```

```
Output of Program
English Sentence: i like apples
POS Translation:
PRON VERB NOUN
```

Result

Thus, a Sequence-to-Sequence model was successfully developed to convert English sentences into their corresponding POS-tag sequences.